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Real-World AI Harm Case Study: Epic Sepsis Model Near-Miss

A documented AI-harm near-miss is the Epic Sepsis Model, an AI algorithm built into Epic's widely used electronic health record system. Designed as an early warning system, it analyzes roughly 80 clinical variables to predict whether hospitalized patients may develop sepsis. Sepsis can become life-threatening if it is not recognized early, so the goal was to help clinicians reassess patients and start treatment sooner. However, when researchers tested the Epic Sepsis Model at Michigan Medicine using data from 27,697 patients and 38,455 hospitalizations, it did not reliably identify which patients would develop sepsis. This was concerning because the Epic Sepsis Model had already been implemented across multiple real hospital settings, so poor performance could affect real clinical decisions and patient outcomes [1].

Researchers found that the model was poor at identifying which patients would develop sepsis, even though it occurred in about 7 out of every 100 patients. Its accuracy score (AUC) was 0.63, which is only slightly better than guessing. This means the model often made mistakes, either missing patients who were deteriorating or alerting clinicians to patients who were not. These errors could create false reassurance in one case and alert fatigue in the other, introducing new safety risks instead of helping clinicians catch sepsis earlier.

The issue was not only weak model performance, but also limited transparency and poor governance. Because the model was privately owned by Epic Systems Corporation, hospitals had limited knowledge of how it worked, what data it was trained on, and whether it would perform safely with their patients. The hospital staff and developers assumed that because the model was used in many places, it would work well everywhere, not anticipating that it could still perform poorly in one local setting. It also failed to fully consider the operational risk of whether alerts would be noticed, trusted, ignored, or acted on too late.

This case matters for my AI-assisted ED triage project because, like the Epic Sepsis Model, my tool would also use information collected at triage to flag high-risk ED patients. It shows that clinical AI should not be trusted just because it comes from a reputable source or it worked somewhere else; it needs to be thoroughly tested in the local ED before it is used for patient care.

The main takeaway is that AI harm can happen when a tool is poorly tested, poorly understood, or poorly fitted to clinical work. For my project, nurses and clinicians should remain the final decision-makers, and the tool should only be used if it proves safe, useful, and understandable locally.

References:

1. Wong A, Otles E, Donnelly JP, Krumm A, McCullough J, DeTroyer-Cooley O, et al. External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients. JAMA Intern Med. 2021 Aug 1;181(8):1065–70. doi:10.1001/jamainternmed.2021.2626

2. Cull J, Brevetta R, Gerac J, Kothari S, Blackhurst D. Epic Sepsis Model Inpatient Predictive Analytic Tool: A Validation Study. Crit Care Explor. 2023 Jun 30;5(7):e0941. doi:10.1097/CCE.0000000000000941 PubMed PMID: 37405252; PubMed Central PMCID: PMC10317482.